

Diffusion: a comparison between liquid and solid polymer LiTFSI electrolytes

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Abstract

From careful analyses of pfg-NMR data, it is demonstrated that the size of the diffusing $\text{Li}^+ \cdot x\text{H}_2\text{O}$ complex in an aqueous solution of LiTFSI is strongly dependent on salt concentration, with the number of solvating water units ranging from six in dilute $(\text{H}_2\text{O})_{500}\text{LiTFSI}$ to two in highly concentrated $(\text{H}_2\text{O})_5\text{LiTFSI}$. Such a relationship is explained by a mass tendency toward a lower solvation number as the number of available H_2O molecules per lithium ion decreases. In a liquid $(\text{PEO})_n\text{LiTFSI}$ system, a contrasting situation prevails, since the size of the diffusing Li^+ complex is almost constant over a large salt concentration range ($5 \leq n \leq 50$). Our interpretations of these data imply that one PEO chain, containing on average nine ether oxygen units, is able to dissolve up to two lithium ions, but exclude the possibility of cationic crosslinks between different PEO chains and direct ionic interactions. For solid $\text{P}(\text{EO})_n\text{LiTFSI}$ electrolytes, a significantly lower value for the diffusion coefficient of the small lithium ions as compared to that of the large TFSI ions ($D_{\text{Li}} = 0.2D_{\text{TFSI}}$) was found for all salt concentrations investigated ($5 \leq n \leq 50$). This observation fits in with recent structural observations, which suggest that lithium ions move as single entities in this specific system and require a rather complicated solvation–desolvation step for long-range motion. In all electrolytes investigated, both liquid and solid, the large and bulky TFSI ion appears to be moving as a single unit, thus manifesting the excellent ionization properties of the LiTFSI salt.

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1. Introduction

Since the initial discovery that polar polymers are able to dissolve salts to form ion conducting solid polymer electrolytes (SPEs), much attention has been devoted to these exciting materials [1]. The proposed technical applications range from rechargeable, envi-

ronment-friendly consumer batteries and electric vehicle propulsion to light-emitting electrochemical cells [2,3]. In spite of these promising possibilities, a required combination of appealing transport properties and structural rigidity appears difficult to achieve, possibly because a relation between the ion transport process and local structure in the conducting amorphous phase has been elusive [4]. With this in mind, several studies have been performed on liquid model polymer electrolyte systems with the ambition of correlating structural parameters and transport properties.

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However, it is unclear to which extent such results can be carried over to SPEs, considering that vehicular solvent–solute transport modes are active in liquids but not in solid materials, e.g., SPEs, but foremost since recent reports support a surprisingly ordered substructure in rubbery SPEs [5–7].

In order to shed some light on this issue, we have measured the ionic self-diffusion coefficients as a function of salt concentration for the salt $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ (LiTFSI) dissolved in three distinctly different solvents, namely solid poly(ethylene oxide) (PEO-S), liquid PEO (PEO-L), and H_2O . PEO is the archetypal polymeric solvent [8] with a rather high solvating power for various alkaline salts despite its low dielectric constant ($\epsilon \approx 5$), and a number of its SPE derivatives exhibit relatively appealing transport properties in the amorphous phase [9,10]. H_2O , on the other hand, has a notably high dielectric constant ($\epsilon=78$), and it is thus expected that direct ion–ion interactions will be effectively screened in the H_2O –LiTFSI system, especially since the delocalized charge of the TFSI anion in itself appears to promote ionization. This feature establishes the H_2O –LiTFSI system as a baseline against which the other systems can be compared, and we use this information, in conjunction with various theoretical models, for an evaluation of the local structure and the diffusion process in the PEO-based systems.

2. Experimental

LiTFSI (a gift from the 3M company) was dried under vacuum at 150°C before use. PEO-S (Polysciences, $M_w=5 \times 10^6$ g/mol) and PEO-L (Polysciences, dimethyl ether end capped PEO, $M_w=400$ g/mol) were used as received. Solid polymer electrolytes were prepared under an inert nitrogen atmosphere by dissolving preweighed amounts of PEO-S and LiTFSI in acetonitrile. The co-solvent was allowed to evaporate, before the samples were further dried under high vacuum for 48 h. The resulting free-standing films were transferred to 5-mm outer diameter NMR tubes, which were subsequently flame-sealed. Liquid electrolytes were prepared under nitrogen by a direct mixing of LiTFSI salt and the H_2O or PEO-L solvent. These samples were stirred until a homogeneous solution was obtained and thereafter transferred to NMR tubes. Samples with ether oxygen-to-lithium

values (n) ranging from 5 to 50 were prepared with solid and liquid PEO as the solvent, respectively, while $\text{H}_2\text{O}/\text{Li}$ ratios of 5 to 500 were synthesized for the water-based samples.

^{19}F , ^7Li and ^1H pfg-NMR diffusion measurements were performed on a Varian CMX Infinity spectrometer, operating at a proton frequency of 400 MHz, using the spin-echo [11] and the stimulated spin-echo [12] pulse sequences. The spectrometer was equipped with a double resonance diffusion probe capable of producing linear gradients up to 3.4 T/m. The 90° pulse lengths were 12 μs (^7Li) and 16 μs (^{19}F and ^1H), respectively, and the waiting time between experiments was typically three to five times longer than the longitudinal relaxation time, T_1 . The temperature (T) was maintained at 85°C for all experiments, and controlled to $\pm 0.5^\circ\text{C}$ by means of a heated air stream passing outside the glass vessel containing the sample. The amplitude of the peaks are attenuated according to the Stejskal–Tanner equation:

$$A = A_0 e^{-\gamma^2 g^2 \delta^2 D(\Delta - \delta/3)} \quad (1)$$

where A and A_0 are the peak amplitudes with and without applied gradients, γ the magnetogyric ratio, g the gradient strength, δ the gradient pulse length and Δ the diffusion time [11]. Typically, 20–40 experiments were performed for each diffusion measurement, during which the gradient strength was varied and all other parameters kept constant, allowing for the peak amplitude to be fitted according to Eq. (1). Several such diffusion measurements were made on each sample with Δ varying between 2.5 and 500 ms. The shorter diffusion times were used for PEO-L and H_2O -based samples to prevent artifacts arising from convection in these low viscosity samples.

3. Results and discussion

In Fig. 1, diffusion data at $T=85^\circ\text{C}$ are presented as a function of salt weight fraction for three different LiTFSI electrolyte systems using H_2O , PEO-L and PEO-S as the solvent, respectively. The general trends are that the magnitudes of the diffusion coefficient of lithium ions (D_{Li}) and TFSI ions (D_{TFSI}) depend strongly on the choice of solvent, and that the values for different electrolytes span a wide range of more

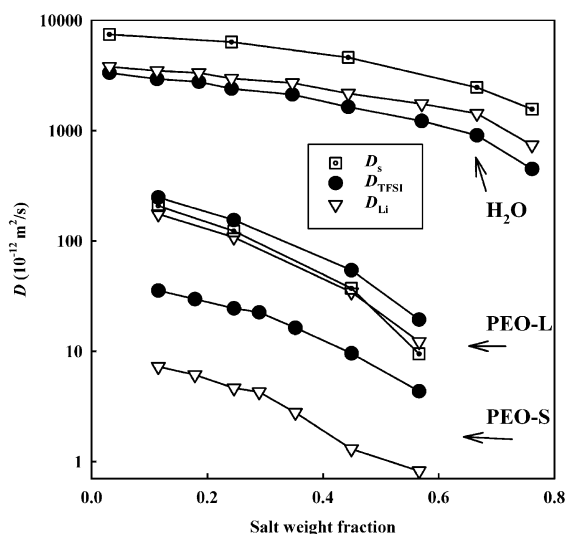


Fig. 1. Self-diffusion coefficients of solvent (D_s), TFSI ions (D_{TFSI}) and lithium ions (D_{Li}) as a function of salt weight fraction in three electrolytes at $T=85^\circ\text{C}$. The solvents in the various systems are from top to bottom: H_2O , liquid PEO, and solid PEO.

than three decades. Furthermore, the interrelationship between the diffusion coefficient of the solvent (D_s), D_{Li} and D_{TFSI} is different in the two liquid systems, whereas the relative deviation between D_{Li} and D_{TFSI} in the solid system is significantly larger than in its liquid counterparts. For the solid system, D_s is effectively equal to zero due to the entangled nature of the PEO chains, which precludes any long-range motion of the solvent. Pfg-NMR data have previously been presented for a similar solid PEO–LiTFSI system ($M_w=900\,000\text{ g/mol}$) in the concentrated regime as a function of T [13]. For overlapping stoichiometries, D_{Li} and D_{TFSI} exhibit similar concentration dependencies although the absolute values are somewhat lower in our study.

In liquid solutions, a diffusion coefficient (D_i) is commonly considered to be a function of T , the viscosity (η), and the effective radius of the diffusing species (R_i), and is routinely analyzed in the context of the classical Stokes–Einstein equation [14,15]:

$$D_i = \frac{kT}{6\pi\eta R_i} \quad (2)$$

where k is the Boltzmann constant. However, this relationship was originally derived for rigid solute spheres diffusing in a continuum of solvent, and Chen

et al. [16] later showed that it breaks down when the solute size is less than five times that of the solvent. None of the diffusing species presented in Fig. 1 fulfills this requirement, and the use of the Stokes–Einstein equation for quantitative analysis is therefore highly doubtful. This is also verified by a direct comparison of the measured and calculated value for the “least unfitting” solute–solvent combination in Fig. 1, namely the dilute aqueous solution of TFSI ions. Here, the calculated value of $D_{\text{TFSI}}=1.96\times 10^{-9}\text{ m}^2/\text{s}$, using $\eta_{\text{H}_2\text{O}}=3.337\times 10^{-4}\text{ kg/m s}$ [17] and $R_{\text{TFSI}}=4.02\times 10^{-10}\text{ m}$ (see below for derivation of this value), is significantly lower than the experimentally obtained $D_{\text{TFSI}}=3.34\times 10^{-9}\text{ m}^2/\text{s}$. Other approaches to find an analytical expression for D_i , developed for cases where the solute and solvent are of a similar magnitude, report minor changes to Eq. (2), but all include the same inverse R_i dependency [18–20]. Subsequently, it appears possible to relate D_i to $1/R_i$ and in the process eliminate the η dependency, as shown below, where subscripts i and j represent two diffusing species in the same system:

$$D_i/D_j = R_j/R_i \quad (3)$$

This procedure was also employed in Ref. [21]. For an ion dissolved in water, the effective radius of the diffusing solute–solvent complex (R_i) can be related to that of the unsolvated ion ($R_{i,0}$) by the equation:

$$(4/3)\pi R_i^3 = (4/3)\pi R_{i,0}^3 + x(V_{\text{H}_2\text{O}}/N_A) \quad (4)$$

in which x is the solvation number of water, $V_{\text{H}_2\text{O}}$ is the molar volume of water and N_A is Avogadro constant [22]. Note that the solvated ion represents the ion and the number of solvent molecules that have lost their own translational freedom and move along with the ion in its haphazard movement in the solution. Unfortunately, the reported value for this solvation number (also called primary hydration number in the case of water) to a certain extent is dependent on the employed measurement method [23]. However, for the thoroughly studied case of lithium ions dissolved in water, there appears to be a consensus for $x=5\pm 1$ [23]. The large size and strongly delocalized charge of the TFSI ion, on the other hand, implies a general solvation number of zero [23,24]. Again, we wish to emphasize that this does not

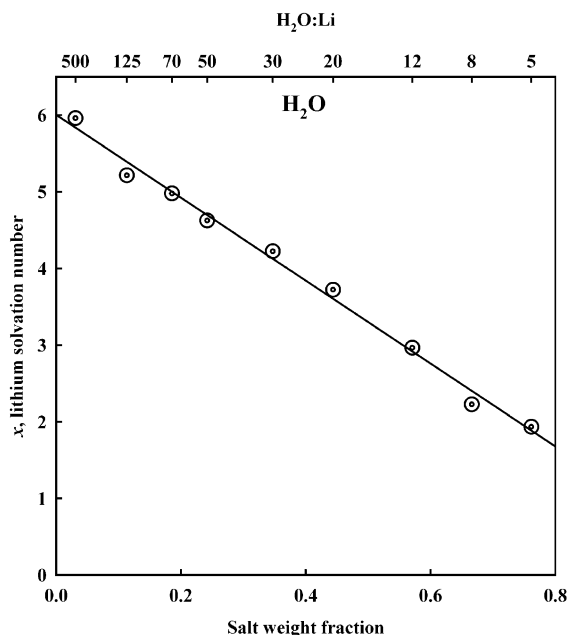


Fig. 2. The lithium ion solvation number (x) as a function of salt weight fraction in H_2O –LiTFSI at $T=85^\circ\text{C}$. The solid line is intended as a guide to the eye only.

suggest that solvent molecules are not in contact with the TFSI ion. They are, but only that the lifetime of the solvent–TFSI interactions, is short in comparison with the time between ionic movements. The high dielectric constant of the H_2O solvent, in conjunction with the high ionization power of the LiTFSI salt, also makes it reasonable to assume that direct ionic interactions are effectively screened in the H_2O –LiTFSI system, and that anions and cation–solvent complexes move as separate entities.

The effective radius of the TFSI ion is a key parameter in the following discussion. We employed Eq. (3), with the most dilute aqueous solution values for D_{TFSI} and $D_{\text{H}_2\text{O}}$ and $R_{\text{H}_2\text{O}}=1.81\text{ \AA}$ [25], to obtain $R_{\text{TFSI}}=4.02\text{ \AA}$. This value is well within range of other reported values of, i.e. $3.26\text{ \AA}$ obtained from crystallographic data [26], and $4.5\text{ \AA}$ extracted from theoretical simulations [24]. The radius of the unsolvated lithium ion, $R_{\text{Li},0}=0.68\text{ \AA}$, is directly available in the literature [27]. With this information, and the experimental data presented in Fig. 1, it is a straightforward task to use Eq. (3) to calculate the size of diffusing Li–solvent complexes ($R_{\text{Li}}=R_{\text{TFSI}}(D_{\text{TFSI}}/D_{\text{Li}})$), and to use Eq. (4)

to translate the aqueous values into a lithium solvation number as presented in Fig. 2. The limiting value of $x=6$ for a dilute H_2O –LiTFSI sample corresponds well with previously reported values in similar systems [23]. The observed trend of a decreasing x with increasing salt weight fraction is also as expected from a mass tendency viewpoint. Note, for instance, that the most concentrated composition in Fig. 2, having an x value of 2, only allows a maximum of five H_2O molecules to interact with each lithium ion, and that such a situation only pertains if there is no free water at all in the solution.

For the liquid $\text{P(EO)}_n\text{LiTFSI}$ system, the results presented in Fig. 3 require a more elaborate discussion. The effective radius of a solvated lithium ion, R_{Li} , calculated using Eq. (3), shows that the average size of Li^+ –PEO complexes is rather large ($R_{\text{Li}}\approx 6\text{ \AA}$) and, more importantly, almost independent of the salt weight fraction in the investigated range. The PEO chains in this specific system contain, on average, nine ether oxygen units, where each unit is capable of interacting with a lithium ion. Computational results for different ether oxygen containing ligands, including PEO, support an ether oxygen solvation number of

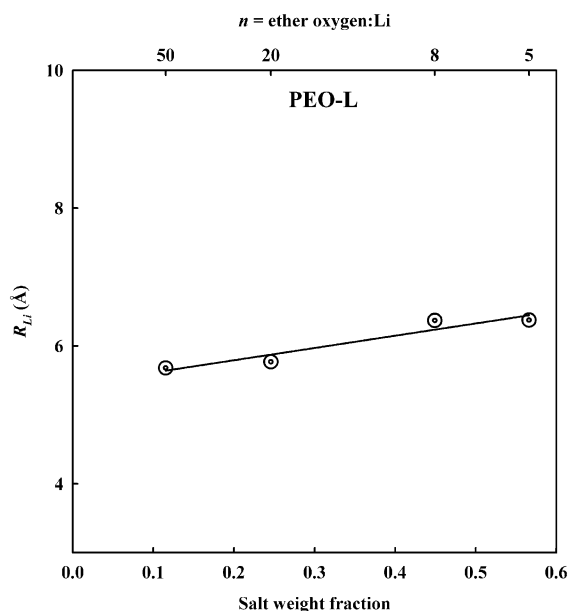


Fig. 3. The effective radius of the solvated lithium ion (R_{Li}) as a function of salt weight fraction in liquid $\text{P(EO)}_n\text{LiTFSI}$ at $T=85^\circ\text{C}$. The solid line is intended as a guide to the eye only.

four to five for lithium ions [28,29]. The consequence, for this specific system in the investigated concentration regime ($5 \leq n \leq 50$), is that one PEO chain is able to dissolve up to two lithium ions, and in the process form a distinct $(\text{Li}^+)_y\text{PEO}$ complex ($y=1, 2$). Such a model fits the experimental data in an excellent way, and the observed minor increase in R_{Li} with increasing salt weight fraction in Fig. 3 matches the expected gradual transition from a mixture of PEO and $(\text{Li}^+)_1\text{PEO}$ chains to a mixture of $(\text{Li}^+)_1\text{PEO}$ and $(\text{Li}^+)_2\text{PEO}$ chains. The very minor dependence of R_{Li} on salt concentration also excludes the possibility of significant lithium ion mediated solvent–solvent interaction, which would correlate to a substantial increase in R_{Li} . Furthermore, the values of D_{Li} and D_{S} (see Fig. 1) show a very minor discrepancy over the entire concentration interval and actually meet at an n value around six, as is expected when all lithium ions and PEO molecules perform diffusive motion collectively. An almost constant $D_{\text{Li}}/D_{\text{TFSI}}$ ratio of 0.7 also reinforces that direct cation–anion interactions are not prevalent in this system. The latter observation is in contrast to results on liquid PEO– LiCF_3SO_3 [30] and liquid poly(propylene oxide)– $\text{NH}_4\text{CF}_3\text{SO}_3$ [31] systems, where a $D_{\text{cation}}/D_{\text{anion}}$ ratio of 1.0 was interpreted in the context of a collective motion of the cationic and anionic species. This fundamental deviation between TFSI and CF_3SO_3 containing electrolytes fits in perfectly with spectroscopic results, which report that there are no bonds between TFSI ions and lithium ions in liquid $\text{P}(\text{EO})_{10}\text{LiTFSI}$, but significant $\text{Li}^+ - \text{CF}_3\text{SO}_3^-$ interactions in liquid $\text{P}(\text{EO})_{10}\text{LiCF}_3\text{SO}_3$ [32].

In the solid $\text{P}(\text{EO})_n\text{LiTFSI}$ system, we encounter a different situation since ions are moving in what is fittingly called an “immobile solvent” in relation to the entangled nature of the PEO chains; one consequence of this being that D_{S} is effectively equal to zero. Recent structural [6,7] and transport studies [9] on this specific system support a rather distinct ionic substructure also in the amorphous phase (crystalline $\text{P}(\text{EO})_6\text{LiTFSI}$ melts at 57°C and crystalline PEO melts at 69°C [33,34]), where lithium ions are coordinated solely to ether oxygens from surrounding PEO chains, and in the process, being completely shielded from direct interactions with TFSI ions. A small but constant $D_{\text{Li}}/D_{\text{TFSI}}$ value of 0.2, over the whole concentration regime in Fig. 1, supports the

notion of an absence of $\text{Li}^+ - \text{TFSI}^-$ bonds in this system. The significantly faster diffusion of large and bulky TFSI ions, in comparison with small lithium ions, also suggests fundamentally different transport mechanisms. Ratner et al. [35,36] proposed that long-range motion of lithium ions require the creation of new solvation sites, which are directly dependent on rather complicated conformational transformations of the polymer. Shi and Vincent [37] have later shown that long-range cationic motion in SPEs is dependent on cations not being too strongly attached to the polymeric chains. “Free” TFSI ions, on the other hand, should only depend on a redistribution of free volume in order to find a suitable void to move into, thus rationalizing the faster anionic diffusion process. It is interesting to note that D_{TFSI} also is significantly larger than D_{Li} in the related poly(propylene oxide)– LiTFSI system [38,39], while $D_{\text{CF}_3\text{SO}_3}$ is almost identical to D_{Li} in poly(propylene oxide)– LiCF_3SO_3 [40]. Again, an observation in line with spectroscopic results reporting significant ionic aggregation in CF_3SO_3 containing SPEs [5] but not in their TFSI counterparts [7].

4. Conclusions

NMR investigations of the mobile species in electrolytes consisting of LiTFSI salt dissolved in H_2O , liquid PEO and solid PEO solvents, respectively, reveal the distinctly different forms in which lithium ions are diffusing. In the H_2O solvent, lithium ions are transported in $\text{Li}^+ \cdot x\text{H}_2\text{O}$ complexes, with a decreasing size as x is smoothly going from six in dilute $(\text{H}_2\text{O})_{500}\text{LiTFSI}$ to two in highly concentrated $(\text{H}_2\text{O})_5\text{LiTFSI}$. For liquid PEO– LiTFSI electrolytes, a single PEO molecule, containing on average nine ether oxygens, is capable of dissolving and transporting up to two lithium ions in distinct $(\text{Li}^+)_y\text{PEO}$ complexes ($y=1, 2$). Such vehicular solute–solvent transport modes are precluded for solid $\text{P}(\text{EO})_n\text{LiTFSI}$ electrolytes, where a long-range motion of lithium ions instead is dependent on the formation of new solvating environments. An observed constant $D_{\text{Li}}/D_{\text{TFSI}}$ ratio of 0.2 over a large salt concentration range ($5 \leq n \leq 50$) fits in with recent structural results, which report that lithium ions are completely shielded from interactions with TFSI ions in this specific solid system. The absence of ionic

aggregation seem to constitute a general trend for both liquid and solid LiTFSI electrolytes, considering that the TFSI ions are moving as single entities in all of the investigated systems.

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References

- [1] B. Scrosati, C.A. Vincent, *MRS Bull.* (2000) 28.
- [2] B. Scrosati, *Nature* 373 (1995) 557.
- [3] Q. Pei, G. Yu, C. Zhang, Y. Yang, A.J. Heeger, *Science* 269 (1995) 1086.
- [4] L. Edman, Recent Research Developments in Electrochemistry, in press.
- [5] R. Frech, S. Chintapalli, P.G. Bruce, C.A. Vincent, *Chem. Commun.* (1997) 157.
- [6] G. Mao, M.-L. Saboungi, D. Price, M.B. Armand, W.S. Howells, *Phys. Rev. Lett.* 84 (2000) 5536.
- [7] L. Edman, *J. Phys. Chem., B* 104 (2000) 7254.
- [8] D. Baril, C. Michot, M. Armand, *Solid State Ionics* 94 (1997) 35.
- [9] L. Edman, M.M. Doeff, A. Ferry, J. Kerr, L.C. De Jonghe, *J. Phys. Chem., B* 104 (2000) 3476.
- [10] M.M. Doeff, L. Edman, S.E. Sloop, J. Kerr, L.C. De Jonghe, *J. Power Sources* 89 (2000) 227.
- [11] E.O. Stejskal, J.E. Tanner, *J. Chem. Phys.* 42 (1965) 288.
- [12] J.E. Tanner, *J. Chem. Phys.* 52 (1970) 2523.
- [13] W. Gorecki, M. Jeannin, E. Belorizky, C. Roux, M. Armand, *J. Phys. Condens. Matter* 7 (1995) 6823.
- [14] G.G. Stokes, *Trans. Camb. Philos. Soc.* 9 (1856) 5.
- [15] A. Einstein, *Ann. Phys.* 17 (1905) 549.
- [16] S.H. Chen, H.T. Davis, D.F. Evans, *J. Phys. Chem.* 75 (1981) 1422.
- [17] R.C. West (Ed.), *CRC Handbook of Chemistry and Physics*, CRC, FL, 1983, p. F-40.
- [18] W. Sutherland, *Philos. Mag.* 9 (1905) 781.
- [19] S. Glasstone, K.J. Laidler, H. Eyring, *Theory of Rate Processes*, McGraw-Hill, NY, 1941, p. 781.
- [20] A.F. Collings, R. Mills, *J. Chem. Soc., Faraday Trans.* 66 (1970) 2761.
- [21] Y. Aihara, K. Sugimoto, W.S. Price, K. Hayamizu, *J. Chem. Phys.* 113 (2000) 1981.
- [22] E.L. Cussler, *Diffusion—Mass Transfer in Fluid Systems*, Cambridge Univ. Press, NY, 1994, p. 173.
- [23] J.O'M. Bockris, A.K.N. Reddy, *Modern Electrochemistry 1—Ionics*, Plenum, NY, 1998, pp. 139–145.
- [24] P. Johansson, S.P. Gejji, J. Tegenfeldt, J. Lindgren, *Electrochim. Acta* 43 (1998) 1375.
- [25] J.O'M. Bockris, A.K.N. Reddy, *Modern Electrochemistry 1—Ionics*, Plenum, NY, 1998, pp. 48–49.
- [26] M. Ue, *J. Electrochem. Soc.* 141 (1994) 3336.
- [27] R.C. West (Ed.), *CRC Handbook of Chemistry and Physics*, CRC, FL, 1983, p. F179.
- [28] R.I. Blint, *J. Electrochem. Soc.* 142 (1995) 696.
- [29] P. Johansson, J. Tegenfeldt, J. Lindgren, *Polymer* 40 (1999) 4399.
- [30] A. Ferry, G. Orädd, P. Jacobsson, *J. Chem. Phys.* 108 (1998) 7426.
- [31] G. Orädd, M. Furlani, A. Ferry, *Solid State Ionics* 136–137 (2000) 457.
- [32] A. Johansson, A. Gogoll, J. Tegenfeldt, *Polymer* 37 (1996) 1387.
- [33] C. Labrèche, I. Lévesque, J. Prud'homme, *Macromolecules* 29 (1996) 7795.
- [34] L. Edman, A. Ferry, M.M. Doeff, *J. Mater. Res.* 15 (2000) 1950.
- [35] S.D. Druger, A. Nitzan, M.A. Ratner, *J. Chem. Phys.* 79 (1983) 3133.
- [36] M.A. Ratner, *Faraday Discuss. Chem. Soc.* 88 (1989) 91.
- [37] J. Shi, C.A. Vincent, *Solid State Ionics* 60 (1993) 11.
- [38] C. Roux, W. Gorecki, J.-Y. Sanchez, E. Belorizky, *Electrochim. Acta* 43 (1998) 1575.
- [39] C. Roux, W. Gorecki, J.-Y. Sanchez, M. Jeannin, E. Belorizky, *J. Phys. Condens. Matter* 8 (1996) 7005.
- [40] S.E. Lindsey, D.H. Whitmore, W.P. Halperin, J.M. Torkelson, *Polym. Prepr.* 30 (1989) 442.